

Everything You Wanted to Know About Modern Sensors



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Various types of sensors are encountered everywhere by us every day. Modern sensors need not be new types of sensors based on new technologies, but surely these are made using modern engineering practices and precision manufacturing. These sensors are used in modern systems like car dashboards, mobile phones, personal computers, tablets, modern telephones, home communication systems and office access systems. Some of these are not sensors but are transducers (for data-to-electrical or electrical-to-electrical conversions) instead.

Modern sensors can be broadly classified into four types: micro-electro-mechanical-system (MEMS) sensors,

touch sensors, haptic sensors and wireless sensors. Of these, radio frequency (RF) wireless sensors and haptic sensors may be deemed as transducers, while MEMS and touch sensors are primarily sensors.

MEMS sensors

These are very small devices whose components are one micrometre to 100 micrometres in size. There are plenty of such devices or sensors that have from very simple structures without any moving elements to complicated and complex structures with many moving particles managed in micro-electronics. At this micrometric scale, the weight of particles (gravitational force) is irrelevant as compared to other forces, like those caused due to the coefficient of linear expansion due to small temperature changes or molecular absorption from surrounding media (like air) creating molecular forces of attraction/repulsion, resulting in bending of tiny cantilevers created by deposition in microelectronics.

An MEMS must have at least one mechanical element or functionality in the component or device. Sensors obeying above-mentioned conditions fall under MEMS.

Mechanical interfacing or functionality of sensors may also be categorised under transducers. There are several microelectronic transducers such as accelerometers, gyroscopes, pressure sensors, humidity sensors and gas sensors.

Gyroscopes and accelerometers.

Gyroscopes are those that are used to calculate the angular motion or displacement presented in the form of digitalised data. Their main principal of working is to measure angular movements with respect to time, that is, axial measurements on X, Y, Z axes.

TABLE I
Some Sensors Available in the Market

Sensor type	Suggested sensors	Range	Company (country)	Cost (in US\$)
Gyrosensor and accelerometer	L3GD20	250/500/2000 lbs $\pm$ 3G	STMicroelectronics (the USA) Analog Devices (the USA)	3.5
	ADXL335			5
Force sensor	FSR A301	0-1000 lbs	Tekscan (the USA)	16
Humidity sensor	HIS-06-N	0-90 per cent RH 20-80 per cent RH	Hokuriku (Japan) Aosong (China)	2
	DHT11			4
Pressure sensor	LPS25H	260-1260hPa	STMicroelectronics (the USA)	2.4